

Corrected structured proof I

GJS Euclidean estimates $\implies$ a cutoff-uniform local moment and physical Hamiltonian gap

Fresh reconstruction of `part_i/main.tex`

16 August 2026

Status. CONDITIONAL. The derivation below is valid from the explicitly listed cutoff-construction facts and the exact estimates in Glimm–Jaffe–Spencer (GJS), Theorems 1.1.7–1.1.8. It incorporates every local repair found in the audit: the dimensionless-coupling reduction, the correct FKN references, signed truncations, the two time-sign cases, the corrected unit-cell count, the missing exponential-moment prefactor, and the restriction $0 \leq \beta < m_1$.

This document proves no part of the GJS cluster expansion itself. That theorem is an import, displayed verbatim by page image in the separate citation dossier.

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1 Objects and exact imports

Let

$$\mathcal{H} = L^2(\mathcal{S}'(\mathbb{R}), d\phi_0), \quad K(g) = H_0 + V(g), \quad H(g) = K(g) - E(g),$$

where $E(g) = \inf \operatorname{spec} K(g)$ and

$$V(h) = \lambda \int_{\mathbb{R}} h(x) : \mathcal{P}(\varphi_0(x)) : dx, \quad \mathcal{P}(\xi) = \sum_{n=0}^{2p'} a_n \xi^n, \quad a_{2p'} > 0.$$

The scalar product is conjugate-linear in its first argument. The cutoffs satisfy $0 \leq g \leq 1$ and have compact support. Write $\Omega_0 = 1$ for the free vacuum.

Imported assumption 1.1 (cutoff Hamiltonian facts). For every such g , $K(g)$ is self-adjoint and bounded below, $E(g)$ is a simple eigenvalue, and its normalized eigenvector Ω_g can be chosen strictly positive $d\phi_0$ -almost everywhere. Consequently

$$c_g := \langle \Omega_g, \Omega_0 \rangle > 0.$$

For every compactly supported real h , the time-zero multiplication operator $V(h)$ is self-adjoint on its maximal domain.

Imported assumption 1.2 (slab FKN input). For a fixed g , the interacting Feynman–Kac–Nelson formula holds on every finite time slab. The correct source locations are Simon, Theorem III.6 for the free FKN formula and Corollary V.14/Theorem V.15 for its interacting form; GRS, Theorems II.14 and II.16, give the same transfer formula. Simon’s Theorem III.12, cited in the audited manuscript, is not an FKN theorem.

Let $d\Phi$ be the Euclidean Gaussian measure with covariance $(-\Delta + m_0^2)^{-1}$. For compactly supported measurable $0 \leq h \leq 1$, put

$$dq_h = Z(h)^{-1} \exp \left[-\lambda \int h(z) : \mathcal{P}(\Phi(z)) : dz \right] d\Phi.$$

GJS define localized sharp-time monomials Q and a norm $\|Q\|_{\text{GJS}}$ in (1.1.7)–(1.1.9). If $N(\Delta)$ is the total degree in a unit square Δ , a localized term with kernel w has norm

$$K_1 \left[\prod_{\Delta} K_2^{2N(\Delta)} N(\Delta)! \right] \|w\|_2,$$

and a nonlocalized polynomial is normed by summing its localized pieces.

Imported assumption 1.3 (CE1: GJS Theorem 1.1.8). There is $C_8 < \infty$, independent of the cutoff h , such that every admissible sharp-time polynomial obeys

$$\left| \int Q(s) dq_h \right| \leq C_8 \|Q(s)\|_{\text{GJS}}.$$

The theorem statement says “uniformly as $h \uparrow 1$ ”; GJS page 629 explicitly states that the estimates of Theorem 1.1.8 are uniform in the space cutoff h .

Imported assumption 1.4 (CE2: GJS Theorem 1.1.7). If Q_1, Q_2 have disjoint localization squares separated by a strip of maximal width d , then

$$\left| \int Q_1 Q_2 dq_h - \int Q_1 dq_h \int Q_2 dq_h \right| \leq C_7 e^{-m_1 d} \|Q_1\|_{\text{GJS}} \|Q_2\|_{\text{GJS}}, \quad (\text{CE2})$$

where $m_1 > 0$ and C_7 are independent of h, Q_1, Q_2 .

Remark 1.5 (no circular gap assumption). Assumptions 1.1–1.4 include simplicity of the cutoff ground state but do not include isolation of its eigenvalue. The gap proved below is therefore not used to obtain the slab limit.

2 The structured proof

Theorem 2.1 (Euclidean estimates imply (M) and (G)). *There is a dimensionless threshold $\varepsilon_{\mathcal{P}} > 0$ such that, if $0 \leq \lambda/m_0^2 < \varepsilon_{\mathcal{P}}$, then the following statements hold for every cutoff g .*

(i) *If h is real, $\|h\|_{\infty} \leq 1$, and $\text{supp } h$ is contained in an interval of length 2, then*

$$\|V(h)\Omega_g\| \leq M, \quad M^2 = 9C_8 K_1 K_2^{8p'} (4p')! \lambda^2 \left(\sum_{n=0}^{2p'} |a_n| \right)^2. \quad (\text{M})$$

(ii)

$$\text{spec } H(g) \subseteq \{0\} \cup [m_1, \infty), \quad (\text{G})$$

and 0 is a simple eigenvalue.

Proof in Lamport form

⟨1⟩ 1 Reduce the GJS smallness hypothesis to $\lambda/m_0^2 < \varepsilon_{\mathcal{P}}$.

⟨2⟩ 1 Fix source-admissible numbers $M_* > 0$ and $\lambda_* > 0$ such that the GJS expansion is valid at mass M_* whenever $0 \leq \lambda' < \lambda_*$.

Because: GJS formulate the construction for sufficiently large mass and sufficiently small coupling. Choosing one admissible pair merely names the constants in that hypothesis.

⟨2⟩ 2 For arbitrary (m_0, λ) set $s = m_0/M_*$ and apply the unitary space-time dilation stated on GJS page 590. It sends

$$(m_0, \lambda) \mapsto (s^{-1}m_0, s^{-2}\lambda) = \left(M_*, \frac{M_*^2}{m_0^2}\lambda\right). \quad (2.1)$$

⟨2⟩ 3 The dilation introduces no change in the coefficients of $\mathcal{P}$ or in Wick ordering.

Because: In two dimensions the scalar field has scaling dimension zero. More explicitly, if $\tilde{\Phi}(y) = \Phi(y/m_0)$, then

$$\mathbb{E}[\tilde{\Phi}(y)\tilde{\Phi}(y')] = C_{m_0}((y - y')/m_0) = C_1(y - y'),$$

by the substitution $p = m_0 q$ in the Fourier integral for C_{m_0} . Equality of all covariance contractions implies equality of the corresponding Wick polynomials. The Jacobian contributes m_0^{-2} to the interaction.

⟨2⟩ 4 Set $\varepsilon_{\mathcal{P}} = \lambda_*/M_*^2$. Then $\lambda/m_0^2 < \varepsilon_{\mathcal{P}}$ implies $s^{-2}\lambda < \lambda_*$, so the transformed model satisfies the source hypothesis.

⟨2⟩ 5 This proves ⟨1⟩1. □

⟨1⟩ 2 Establish the exact slab transfer identities without assuming a gap.

Fix g and define

$$h_{T,g}(t, x) = \mathbf{1}_{[-T, T]}(t)g(x), \quad \psi_a = e^{-aH(g)}\Omega_0, \quad \hat{\psi}_a = \psi_a/\|\psi_a\|.$$

⟨2⟩ 1 For bounded nonnegative time-zero functions F_1, F_2 , FKN gives, for $0 \leq t \leq 2T$ and $|s| \leq T$,

$$\int F_1(\Phi(-t/2))F_2(\Phi(t/2)) dq_{h_{T,g}} = \frac{\langle F_1\psi_{T-t/2}, e^{-tH(g)}F_2\psi_{T-t/2} \rangle}{\|\psi_T\|^2}, \quad (2.2)$$

$$\int F_1(\Phi(s)) dq_{h_{T,g}} = \frac{\langle \psi_{T-s}, F_1\psi_{T+s} \rangle}{\|\psi_T\|^2}. \quad (2.3)$$

Because: The unshifted formula contains $K(g)$ and the same factor $e^{-2TE(g)}$ in numerator and partition function. Multiplying every semigroup by the corresponding $e^{aE(g)}$ replaces $K(g)$ by $H(g)$ and cancels that factor exactly.

⟨2⟩ 2 Equations (2.2)–(2.3) hold for bounded signed functions.

Because: Write $F = F^+ - F^-$ and expand bilinearly. Each of the four terms is a finite nonnegative FKN integral, so linearity is legitimate.

⟨2⟩ 3 They extend to the unbounded functions used below.

Because: For a real multiplication function F , set $F^{[M]} = (-M) \vee (F \wedge M)$. Apply the bounded identity to $F^{[M]}$. Whenever $F\psi_a \in \mathcal{H}$, $F^{[M]}\psi_a \rightarrow F\psi_a$ by dominated convergence, because $|F - F^{[M]}|^2|\psi_a|^2 \leq |F|^2|\psi_a|^2$. On path space, use $|F_1 F_2| \leq (F_1^2 + F_2^2)/2$ after translating both factors to their respective time slices. The CE1 estimates below supply the required square integrability.

$\langle 2 \rangle 4$ This proves $\langle 1 \rangle 2$. □

$\langle 1 \rangle 3$ Prove the slab vectors converge to the cutoff ground state using simplicity only.

Let $P_g = |\Omega_g\rangle\langle\Omega_g|$. Positivity gives $P_g\Omega_0 = c_g\Omega_g$ with $c_g > 0$. Hence

$$\psi_a = c_g\Omega_g + e^{-aH(g)}(1 - P_g)\Omega_0. \quad (3.1)$$

$\langle 2 \rangle 1$ The second term of (3.1) converges to zero in norm.

Because: If μ is the spectral measure of $(1 - P_g)\Omega_0$, then

$$\|e^{-aH(g)}(1 - P_g)\Omega_0\|^2 = \int_{(0,\infty)} e^{-2aE} d\mu(E) \rightarrow 0$$

by dominated convergence. Simplicity says precisely that $\mu(\{0\}) = 0$; no positive distance from zero is used.

$\langle 2 \rangle 2$ Consequently

$$\hat{\psi}_a \rightarrow \Omega_g, \quad \|\psi_a\| \downarrow c_g, \quad 1 \leq \frac{\|\psi_b\|}{\|\psi_a\|} \leq c_g^{-1} \quad (0 \leq b \leq a), \quad (3.2)$$

and every ratio in (3.2) tends to 1 when both indices tend to infinity with bounded difference.

Because: $e^{-aH(g)}$ is a contraction and $\|\Omega_0\| = 1$.

$\langle 2 \rangle 3$ This proves $\langle 1 \rangle 3$. □

$\langle 1 \rangle 4$ Derive the local interaction moment (M).

Let h satisfy the hypotheses of (M). In Euclidean notation define

$$Q_V = \lambda^2 \sum_{n,n'=0}^{2p'} a_n a_{n'} \iint h(x)h(y) : \Phi(0,x)^n :: \Phi(0,y)^{n'} : dx dy. \quad (4.1)$$

$\langle 2 \rangle 1$ Q_V is exactly the multiplication function $V(h)^2$.

Because: Expanding the square of $\lambda \sum_n a_n \int h(x) : \Phi(0,x)^n : dx$ gives (4.1). No Wick reordering of the product is performed or required; GJS (1.1.5) permits products of individually Wick-ordered powers.

$\langle 2 \rangle 2$ The support of h meets at most three half-open unit spatial cells in positive measure.

Because: An interval of length 2 meets at most three members of the partition $[j, j+1)$ except possibly in endpoints, which have zero Lebesgue measure.

$\langle 2 \rangle 3$ For one ordered pair of cells and degrees n, n' , the GJS local weight is bounded by $K_1 K_2^{8p'} (4p')!$.

Because: The total local degree is at most $n + n' \leq 4p'$. If both variables belong to one square, the weight is $K_1 K_2^{2(n+n')}(n+n')!$. If they belong to two squares, it is $K_1 K_2^{2n} n! K_2^{2n'} n'!$. In both cases

$$K_2^{2(n+n')} \leq K_2^{8p'}, \quad n!n'! \leq (n+n')! \leq (4p')!,$$

after increasing K_2 to at least 1, as GJS do.

⟨2⟩ 4 Summing the at most $3^2 = 9$ ordered cell pairs and all coefficients gives

$$\|Q_V\|_{\text{GJS}} \leq 9K_1K_2^{8p'}(4p')!\lambda^2 \left(\sum_{n=0}^{2p'} |a_n| \right)^2 = M^2/C_8. \quad (4.2)$$

⟨2⟩ 5 For every $T > 0$,

$$\|V(h)\hat{\psi}_T\|^2 = \int V(h)^2 dq_{hT,g} \leq C_8\|Q_V\|_{\text{GJS}} \leq M^2. \quad (4.3)$$

Because: Apply (2.2) at $t = 0$ with the two individually nonnegative functions $F_1 = F_2 = |V(h)|$. Their product is $V(h)^2 = Q_V$. Then apply CE1 and (4.2). This is the required correction to the manuscript's invalid phrase " $F_1F_2 = V(h)^2 \geq 0$ ", which did not specify nonnegative F_1, F_2 .

⟨2⟩ 6 Pass (4.3) to Ω_g .

Because: Choose $T_k \rightarrow \infty$ such that $\|V(h)\hat{\psi}_{T_k}\|$ approaches its liminf. By (4.3), a subsequence converges weakly to some y . The graph of the closed closed linear operator $V(h)$ is a closed linear subspace of $\mathcal{H} \oplus \mathcal{H}$, hence is weakly closed. Since $\hat{\psi}_{T_k} \rightarrow \Omega_g$ strongly, (Ω_g, y) lies in the graph. Thus $y = V(h)\Omega_g$ and weak lower semicontinuity gives

$$\|V(h)\Omega_g\| \leq \liminf_k \|V(h)\hat{\psi}_{T_k}\| \leq M.$$

⟨2⟩ 7 This proves (M). □

⟨1⟩ 5 Obtain correct GJS norm bounds for polynomial time-zero vectors.

Let $Q = \prod_{i=1}^r \varphi_0(f_i)$ with real $f_i \in C_c^\infty(\mathbb{R})$, all supported in an interval I of length L . Let

$$N_I := \#\{j : |I \cap [j, j+1)| > 0\}.$$

⟨2⟩ 1 $N_I \leq \lfloor L \rfloor + 2 \leq L + 2$.

Because: If $j_{\min}$ and $j_{\max}$ are the extreme intersecting cells, then $j_{\max} - j_{\min} - 1 < L$, hence the number $j_{\max} - j_{\min} + 1$ is at most $\lfloor L \rfloor + 2$. The manuscript's bound $L + 1$ is false for nonintegral L ; an interval of length 0.2 can meet two cells.

⟨2⟩ 2 Decomposing each f_i by the unit cells yields

$$\|Q\|_{\text{GJS}} \leq K_1K_2^{2r}r!N_I^{r/2} \prod_{i=1}^r \|f_i\|_2, \quad (5.1)$$

$$\|Q^2\|_{\text{GJS}} \leq K_1K_2^{4r}(2r)!N_I^r \prod_{i=1}^r \|f_i\|_2^2 =: \mathbf{n}(Q)^2. \quad (5.2)$$

Because: Expand the kernel into cell assignments. For any assignment, the product of local factorials is at most the factorial of total degree, and the K_2 exponents sum to twice that degree. For each i ,

$$\sum_{j: I \cap [j, j+1) \neq \emptyset} \|f_i \mathbf{1}_{[j, j+1)}\|_2 \leq N_I^{1/2} \|f_i\|_2$$

by Cauchy–Schwarz. Multiplying these r estimates proves (5.1); applying it to the $2r$ factors in Q^2 proves (5.2).

⟨2⟩ 3 The norm is independent of the common sharp time s .

Because: At every s all factors occupy one temporal row. Changing s changes only that common row label; the spatial cell assignments, local degrees, kernel norms, and product of weights are unchanged. This is the precise statement; arbitrary time translation does not literally preserve the fixed lattice squares.

⟨2⟩ 4 This proves ⟨1⟩5. □

⟨1⟩ 6 Transfer CE1 and CE2 to connected cutoff-ground-state semigroup bounds.

⟨2⟩ 1 CE1 and (2.2) at $t = 0$ imply

$$\sup_T \|Q\hat{\psi}_T\|^2 \leq C_8 \mathfrak{n}(Q)^2. \quad (6.1)$$

Because: Apply the signed-truncation construction of ⟨2⟩3 to Q ; equivalently use $F_1 = F_2 = |Q|$. Then the Euclidean integrand is Q^2 , to which CE1 and (5.2) apply. The weak-graph argument of ⟨1⟩4 also gives $\Omega_g \in D(Q)$.

⟨2⟩ 2 For fixed $s \in \mathbb{R}$,

$$\int Q(s) dq_{h_{T,g}} \longrightarrow \langle \Omega_g, Q\Omega_g \rangle. \quad (6.2)$$

Because: For $T > |s|$, (2.3) gives

$$\int Q(s) dq_{h_{T,g}} = \langle \hat{\psi}_{T-s}, Q\psi_{T+s}/\|\psi_T\| \rangle \frac{\|\psi_{T-s}\|}{\|\psi_T\|}. \quad (6.3)$$

The first vector tends strongly to Ω_g , and the last factor tends to 1. Write the middle vector as

$$\frac{Q\psi_{T+s}}{\|\psi_T\|} = \frac{\|\psi_{T+s}\|}{\|\psi_T\|} Q\hat{\psi}_{T+s}. \quad (6.4)$$

If $s \geq 0$, the scalar ratio in (6.4) is at most 1. If $s < 0$, it is at most c_g^{-1} by (3.2). Thus (6.4) is bounded in either case. Every weak subsequential limit equals $Q\Omega_g$ by weak closedness of the graph and because the scalar ratio tends to 1. Hence the whole middle vector converges weakly to $Q\Omega_g$, proving (6.2). This separate treatment of $s < 0$ repairs the false inequality $\|\psi_{T+s}\| \leq \|\psi_T\|$ used for both signs in the manuscript.

⟨2⟩ 3 For $t \geq 0$,

$$\langle Q\Omega_g, e^{-tH(g)} Q\Omega_g \rangle \leq \liminf_{T \rightarrow \infty} \int Q(-t/2) Q(t/2) dq_{h_{T,g}}. \quad (6.5)$$

Because: With $u_T = Q\psi_{T-t/2}/\|\psi_T\|$, the signed-truncation extension of (2.2) gives the exact equality

$$\int Q(-t/2) Q(t/2) dq_{h_{T,g}} = \|e^{-tH(g)/2} u_T\|^2. \quad (6.6)$$

The ratio bound (3.2) and (6.1) make u_T bounded. Write $u_T = (\|\psi_{T-t/2}\|/\|\psi_T\|) Q\hat{\psi}_{T-t/2}$; the scalar tends to 1 and is uniformly bounded. From every subsequence choose a further one on which u_T converges weakly. Since $\hat{\psi}_{T-t/2} \rightarrow \Omega_g$ strongly, weak closedness of the graph of the closed operator Q identifies that weak limit as $Q\Omega_g$. Thus the full net converges weakly to $Q\Omega_g$. Boundedness of $e^{-tH/2}$ and weak lower semicontinuity of the norm prove (6.5).

⟨2⟩ 4 For $t > 2$, CE2 implies

$$\left| \int Q(-t/2) Q(t/2) dq_h - \int Q(-t/2) dq_h \int Q(t/2) dq_h \right| \leq C_7 e^{2m_1} e^{-m_1 t} \|Q\|_{\text{GJS}}^2. \quad (6.7)$$

Because: Write the source-prescribed finite localization decomposition as $Q = \sum_{a \in A} Q_a$, so that $\|Q\|_{\text{GJS}} = \sum_a \|Q_a\|_{\text{GJS}}$. The two temporal rows of every pair $Q_a(-t/2), Q_b(t/2)$ are separated by a horizontal strip of width at least $t - 2$. Therefore CE2 and the triangle inequality give

$$\begin{aligned} |\text{Cov}_{dq_h}(Q(-t/2), Q(t/2))| &\leq \sum_{a,b} |\text{Cov}_{dq_h}(Q_a(-t/2), Q_b(t/2))| \\ &\leq C_7 e^{-m_1(t-2)} \sum_{a,b} \|Q_a\|_{\text{GJS}} \|Q_b\|_{\text{GJS}} \\ &= C_7 e^{2m_1} e^{-m_1 t} \|Q\|_{\text{GJS}}^2. \end{aligned}$$

This decomposition is required because GJS Theorem 1.1.7 is stated for localized monomials; applying it directly to a general Q would be invalid.

$\langle 2 \rangle 5$ Let $\psi_Q = Q\Omega_g - \langle \Omega_g, Q\Omega_g \rangle \Omega_g$. Then

$$\langle \psi_Q, e^{-tH(g)} \psi_Q \rangle \leq \hat{C}_Q e^{-m_1 t} \quad (t \geq 0), \quad (6.8)$$

where

$$\hat{C}_Q = \max\{C_7 e^{2m_1} \|Q\|_{\text{GJS}}^2, e^{3m_1} \|\psi_Q\|^2\}.$$

Because: For $t > 2$, expand the connected expression, use (6.5), then (6.7), and use (6.2) for both one-point terms. Explicitly, if a_T is the connected Euclidean difference and b_T the product of one-point functions, then

$$\liminf_T (a_T + b_T) \leq \sup_T |a_T| + \lim_T b_T.$$

For $0 \leq t \leq 3$, positivity of $H(g)$ gives $\langle \psi_Q, e^{-tH(g)} \psi_Q \rangle \leq \|\psi_Q\|^2 \leq e^{3m_1} \|\psi_Q\|^2 e^{-m_1 t}$.

$\langle 2 \rangle 6$ This proves $\langle 1 \rangle 6$. □

$\langle 1 \rangle 7$ Prove polynomial vectors are dense.

Define $d\mu_g = |\Omega_g|^2 d\phi_0$. Since $\Omega_g > 0$ almost everywhere, multiplication by Ω_g is a unitary map from $L^2(d\mu_g)$ to $\mathcal{H}$. It therefore suffices to prove that cylinder polynomials are dense in $L^2(d\mu_g)$.

$\langle 2 \rangle 1$ Every real cylinder variable $X_f = \varphi_0(f)$ has a nonzero exponential-moment radius.

Because: Let f have support in an interval of length L , set $K' = K_2^2 N_I^{1/2} \|f\|_2$, and apply (6.1) to $Q = X_f^n$. Then

$$\mathbb{E}_{\mu_g}[X_f^{2n}] \leq C_8 K_1 (K')^{2n} (2n)!. \quad (7.1)$$

Since $e^{s|x|} \leq 2 \cosh(sx)$,

$$\mathbb{E}_{\mu_g} e^{s|X_f|} \leq 2 \max\{1, C_8 K_1\} \sum_{n=0}^{\infty} (sK')^{2n} < \infty \quad (0 \leq s < K'^{-1}). \quad (7.2)$$

The factor K_1 in (7.2), omitted in the manuscript, affects only the prefactor, not the radius.

$\langle 2 \rangle 2$ If $\zeta \in L^2(d\mu_g)$ is orthogonal to all cylinder polynomials, then its conditional expectation on every finite cylinder sigma-algebra is zero.

Because: Fix real $e_1, \dots, e_k$ and define

$$F(z_1, \dots, z_k) = \mathbb{E}_{\mu_g} \left[\bar{\zeta} \exp \left(\sum_{j=1}^k z_j X_{e_j} \right) \right].$$

By Cauchy–Schwarz, Hölder, and (7.2), F is analytic in a nonempty polystrip about $i\mathbb{R}^k$. Differentiation under the integral is justified by the same exponential dominator. Every derivative at zero

is an inner product with a polynomial, hence is zero. Repeated application of the one-variable identity theorem gives $F = 0$ throughout the connected polystrip, in particular on $i\mathbb{R}^k$. Thus the finite complex measure

$$B \mapsto \mathbb{E}_{\mu_g} [\bar{\zeta} \mathbf{1}_{\{(X_{e_1}, \dots, X_{e_k}) \in B\}}]$$

has zero Fourier transform. Uniqueness of Fourier transforms of finite measures gives $\mathbb{E}_{\mu_g} [\zeta \mid \sigma(X_{e_1}, \dots, X_{e_k})] = 0$.

⟨2⟩ 3 A countable increasing family of these finite sigma-algebras exhausts the cylinder sigma-algebra.

Because: Choose (e_j) dense in real $L^2(\mathbb{R})$. The free covariance $C = (2\sqrt{-\partial_x^2 + m_0^2})^{-1}$ is bounded, so

$$\mathbb{E}_{\phi_0} |X_f - X_e|^2 = \langle f - e, C(f - e) \rangle \leq \|C\| \|f - e\|_2^2.$$

For an approximating sequence, a subsequence converges ϕ_0 -almost surely and hence μ_g -almost surely because $\mu_g \ll \phi_0$. Therefore every X_f is measurable with respect to $\bigvee_k \sigma(X_{e_1}, \dots, X_{e_k})$ modulo null sets.

⟨2⟩ 4 The martingale convergence theorem now gives $\zeta = 0$.

Because: The conditional expectations in ⟨2⟩2 are all zero and converge in L^2 to the conditional expectation on the exhausted cylinder sigma-algebra, which is ζ itself. Hence cylinder polynomials, and therefore $\text{span}\{Q\Omega_g\}$, are dense.

⟨2⟩ 5 This proves ⟨1⟩7. □

⟨1⟩ 8 Deduce the physical Hamiltonian gap (G).

Fix $0 \leq \beta < m_1$ and let $P_\beta = \mathbf{1}_{[0, \beta]}(H(g))$.

⟨2⟩ 1 For every centered polynomial vector ψ_Q , $P_\beta \psi_Q = 0$.

Because: The spectral theorem and (6.8) give, for every $t \geq 0$,

$$e^{-\beta t} \|P_\beta \psi_Q\|^2 \leq \langle \psi_Q, e^{-tH(g)} \psi_Q \rangle \leq \hat{C}_Q e^{-m_1 t}.$$

Multiply by $e^{\beta t}$ and let $t \rightarrow \infty$. Since $m_1 - \beta > 0$, $\|P_\beta \psi_Q\| = 0$.

⟨2⟩ 2 $P_\beta \Omega_g = \Omega_g$.

Because: $H(g)\Omega_g = 0$ and $0 \in [0, \beta]$.

⟨2⟩ 3 $P_\beta = |\Omega_g\rangle\langle\Omega_g|$.

Because: By ⟨1⟩7, the complex span of Ω_g and the centered polynomial vectors is dense. Steps ⟨2⟩1–⟨2⟩2 show that P_β maps this dense subspace into $\mathbb{C}\Omega_g$. Continuity of the bounded projection extends this containment to all of $\mathcal{H}$, while $P_\beta \Omega_g = \Omega_g$ shows its range is exactly that line.

⟨2⟩ 4 There is no spectrum in $(0, m_1)$.

Because: If an open interval $J \subseteq (0, m_1)$ had a nonzero spectral projection, choose $\beta < m_1$ above its right endpoint. Then $\mathbf{1}_J(H(g)) \leq P_\beta - |\Omega_g\rangle\langle\Omega_g| = 0$, a contradiction. Thus $\text{spec } H(g) \subseteq \{0\} \cup [m_1, \infty)$; simplicity was imported in Assumption 1.1.

⟨2⟩ 5 This proves (G), hence Theorem 2.1. □

3 Audit map back to the manuscript

Location	Defect	Why the written step fails	Exact correction above
lines 305, 536	wrong source	Simon III.12 is a projection-product estimate, not FKN.	Assumption 1.2; use Simon III.6 and V.14–V.15 or GRS II.14, II.16.
line 352	positivity hypothesis unstated	The slab lemma requires each $F_i \geq 0$, not merely $F_1 F_2 \geq 0$.	$\langle 1 \rangle 4 \langle 2 \rangle 5$: $F_1 = F_2 = V(h) $.
lines 376–377	false cell count	A length- L interval can meet more than $L + 1$ unit cells.	$\langle 1 \rangle 5$: use $N_I \leq \lfloor L \rfloor + 2 \leq L + 2$.
lines 404–420	incomplete unbounded passage	Signed observables are outside the stated nonnegative FKN lemma.	$\langle 1 \rangle 2 \langle 2 \rangle 2 - \langle 2 \rangle 3$ and $\langle 1 \rangle 6$: signed truncations plus an explicit L^2 dominator.
line 412	false for $s < 0$	$\ \psi_{T+s}\ \leq \ \psi_T\ $ holds only for $s \geq 0$.	$\langle 1 \rangle 6 \langle 2 \rangle 2$: split the two signs and use the bound c_g^{-1} .
line 455	missing factor	$C_8 K_1$ cannot disappear from the moment series.	$\langle 1 \rangle 7 \langle 2 \rangle 1$: keep it outside the geometric series.
line 484	missing range restriction	$P_{[0,\beta]} \Omega_g = \Omega_g$ is false for $\beta < 0$.	$\langle 1 \rangle 8$: state $0 \leq \beta < m_1$.
standing assumption	scaling suppressed	The source smallness conditions are dimensional until scaling is written.	$\langle 1 \rangle 1$: exact dilation and definition of $\varepsilon_{\mathcal{P}}$.

4 Logical conclusion

The argument establishes (M) and (G) for every cutoff g , with constants independent of g , conditional on the enumerated construction facts and the GJS estimates. None of the identified defects forces a change in the claimed gap m_1 or in the qualitative local-moment conclusion; each is repaired explicitly above. The proof does not establish the GJS estimates, cutoff-Hamiltonian self-adjointness, ground-state simplicity, or FKN; those remain imported theorems and must not be hidden by the phrase “conditional only on GJS Theorems 1.1.7–1.1.8.”